

Types of Probability

I. Theoretical / Classical

A procedure or experiment does not need to be executed in order to determine the probabilities of the outcomes because each outcome is equally likely.

For example: Toss a six-sided die and the probability of obtaining a 6 is $1/6$.



Rule: Probability = (number of ways to obtain a particular outcome) ÷ (total number of outcomes)

NOTICE YOU DID NOT HAVE TO Toss the Die in what is the order to answer prob. of getting an even # when tossing a die? $3/6$

II. Empirical / Relative Frequency

A procedure that must be performed to determine the probabilities. Counting must be performed and the relative frequency of outcomes must be determined.

For example: Pick one piece of candy from a 1 pound bag of candy. What is the probability that candy will be blue?



Rule: Probability = (number of blue candies) ÷ (total number of candies)

Therefore: We would have to count to answer this question.

III. Subjective

A probability that is based on professional opinion or an educated guess.



For example: What is the probability of a meteor hitting downtown Chicago? Turns out a meteor shower was over Chicago just a few days ago. Predicting a meteor shower is based on fact; predicting it will hit Chicago would be an educated guess.

Rule: None other than the use of Science to determine likelihood of certain events.

also, personal opinion (Debbie usu. calls me on MONDAY)

Today's Chicago weather: 15% chance of Rain

Law of Large Numbers

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► Special Note re Theoretical vs. Empirical Probabilities

CLASSICAL / Relative Frequency

Without needing to toss a coin, we know the probability of heads is $1/2$ and the probability of tails is $1/2$.

Theoretically, if we toss the coin 20 times, we expect heads 10 times and tails 10 times.

However, if we actually toss the coin, i.e., we are now considering the Empirical probabilities of coin tosses, we could toss the coin 20 times and get 17 heads and 3 tails.

For 20 tosses of coin:

Theoretical Probability 50% heads 50% tails ✓

Empirical Probability 85% heads 15% tails (as an example) ✓

The Law of Large Numbers

As an experiment is repeated a large number of times, the Empirical probabilities approach the Theoretical probabilities.

For 1,000,000 tosses of a coin:

Theoretical Probability 50% heads 50% tails

Empirical Probability 50% heads 50% tails

→ I'm tossing the coin

Notation

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Notation

✓ Let x denote a particular event

we could name the event A , B , C , ..., but never P , S , or Z as these have special meanings in Statistics *safe to use x*

Then $P(x)$ is the probability of event x occurring

Example

*in alg., x is a number
in prob., x is an event*

Let $x =$ obtain an even number less than 5 when rolling a die

Or, let $x = 2$ or $x = 4$ $x =$ obtaining a 2 or a 4

Then $P(x) = 2/6 = 1/3$ because there are 2 sides that have even numbers less than 5

It could also be written as $P(x) = 33.3\%$

1. let $x =$ describe the event
2. $P(x) =$ probability % or Fraction or a decimal

The Numbers

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- ▶ All probabilities are between 0 and 1 inclusive (not $<$, but $\leq$)

$$0 \leq P(x) \leq 1 \quad \text{or} \quad 0\% \leq P(x) \leq 100\%$$

- ▶ The sum of all probabilities of a procedure is one

$$\sum P(x) = 1 \quad \text{or} \quad \sum P(x) = 100\%$$

Σ greek capital sigma
it means "Add"

When we toss a die, the probability of each side is $1/6$ and since there are 6 sides, $1/6$ times 6 = 1 = 100%

- ▶ The probability of an impossible event is zero

$$P(\text{an impossible event}) = 0 = 0\%$$

$$P(\text{you can walk on the ceiling}) = 0 = 0\%$$

- ▶ The probability of a certain event is one

$$P(\text{a certain event}) = 1 = 100\%$$

$$P(\text{a spider can walk on the ceiling}) = 1 = 100\%$$

→ VERY UNLIKELY

- ▶ The probability of an improbable event is extremely close to zero

$$P(\text{an improbable event}) \leq 0.0001 = 0^+ \quad \leftarrow$$

$$P(\text{you will win a lottery}) = 0.0001 = 0^+$$

because the actual odds of winning might be 1:15,890,700 or 0.00000006293

- ▶ The probability of an unusual event is no more than 5%

$$P(\text{an unusual event}) \leq 5\%$$

$$P(\text{this medicine will have a risk}) \leq 5\%$$

✓ This says it would be very unusual to have go wrong.

- ▶ The probability of a usual event is greater than 5%

$$P(\text{a usual event}) > 5\%$$

$$P(\text{a person in good health will wake up today}) > 5\%$$

SIMPLE PROBABILITY

Probability is the likelihood, or chance, of something happening or not happening.

OR percent

Probability is a ratio. It compares the number of favorable outcomes to the number of possible outcomes. A favorable outcome is the event you are analyzing.

$$\text{probability of an event} = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}}$$

Probability is often written as a percent.

- A probability of 0% means than an event cannot happen.
- A probability of 100% means than an event is certain to happen.

WRITING PROBABILITY AS A NUMBER

Example What is the probability of rolling a 6 using a regular 6-sided die?

Step 1

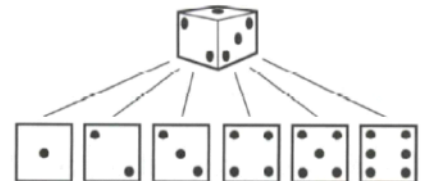
Count outcomes. Since the die has 6 sides, the total number of possible outcomes is 6. Since there is only 1 side with a 6, the number of favorable outcomes is 1.

Step 2

Write the ratio.

$$\text{probability of rolling a 6} = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}} = \frac{1}{6}$$

The probability of rolling a 6 is $\frac{1}{6}$ or 1 chance in 6 or about 17%.



Each face has an equal chance of turning up.

Let x = Roll a 6
 $P(x) = \frac{1}{6}$

Complements & Odds

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The complement of an event occurring is the event not occurring.

So, if $P(x)$ is the probability of event x occurring
Then $P(\bar{x})$ is the probability of event x not occurring

We call " $\bar{x}$ " x -complement.

Note, $\bar{x}$ is the same notation as is used for the mean of a set of quantitative data, but then it is called " x -bar" and we differentiate the two by context.

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You can also figure out the probability of something *not* happening.

Example Oscar has 10 pairs of socks. Of the pairs, 3 are black, 2 are gray, 3 are brown, and 2 are blue. Without looking, Oscar reaches into the drawer and draws out a pair of socks. What is the chance that the socks are *not* blue?

The chance of getting blue is $\frac{2}{10}$, which equals $\frac{1}{5}$, or 20%.

To find the chance of *not* getting blue, subtract from 100%. $100\% - 20\% = 80\%$

Finding the **odds** of an event happening is different from finding probability. To find the odds, write a ratio comparing the number of positive outcomes to the number of negative outcomes. The odds of Oscar drawing out a pair of blue socks are 2 blue to 8 non-blue, or 1 to 4.

Definition: Odds is the ratio of favorable to unfavorable outcomes, written as $a:b$.

$$P(\text{favorable outcomes } a) = P(a) / P(a+b)$$

$$P(\text{unfavorable outcomes } b) = P(b) / P(a+b)$$

$$P(\text{unfavorable outcomes } b) = P(\bar{a}) \text{ in this case, we are using the complement}$$

The odds of Blue socks to the other colors:

2:8

2:8

$$P(2) = 2 / (2+8)$$

$$= 2/10$$

$$P(8) = 8 / (2+8)$$

$$= 8/10$$

Let x = socks are not blue

$$P(x) = 8/10$$

alternate solution

$$\begin{aligned} P(x) &= 1 - P(\text{blue socks}) \\ &= 1 - 2/10 \\ &= 8/10 \end{aligned}$$

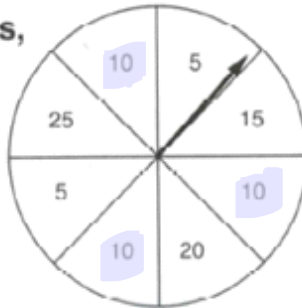
Simple Probability - Examples

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same thing

Solve as directed. For these problems, write probability as a fraction. Express chance as a percent.

A spinner has 8 equal sections, numbered as shown.



1. If you spin the spinner, what is the probability that you will land on a number greater than 12?

let $X = \text{a number} > 12$
 $= 25 \text{ or } 20 \text{ or } 15$

$$P(X) = \frac{3}{8} \\ = 37.5\%$$

2. What is the chance of the spinner landing on 20 or 25?

let $X = 20 \text{ or } 25$

$$P(X) = \frac{2}{8} = 25\%$$

3. What is the probability of *not* landing on 10?

let $X \neq 10$

$$P(X) = 1 - \frac{3}{8} = \frac{5}{8} = 62.5\%$$

Simple Probabilities - Examples, cont'd...

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SUIT	ACE	2	3	4	5	6	7	8	9	10	JACK	QUEEN	KING
CLUBS	A	2	3	4	5	6	7	8	9	10	J	Q	K
DIAMONDS	A	2	3	4	5	6	7	8	9	10	J	Q	K
HEARTS	A	2	3	4	5	6	7	8	9	10	J	Q	K
SPADES	A	2	3	4	5	6	7	8	9	10	J	Q	K

A deck has 52 cards in all. There are 4 suits (diamonds, clubs, spades, and hearts). Each suit has 13 cards: an ace, a king, a queen, a jack, and the numbers 2 through 10.



example:

let $x = \text{draw a spade}$

$$P(x) = 13/52$$

let $x = \text{draw an ACE}$

$$P(x) = \frac{\# \text{ ACEs}}{\text{total \# cards}} = \frac{4}{52} = \frac{1}{13}$$

4. If a card is selected at random, what is the chance of choosing a diamond?

let $x = \text{Diamond}$. $P(x) = 13/52$

5. What is the probability of selecting an ace at random?

let $x = \text{Ace}$. $P(x) = 4/52 = 1/13$

6. What is the probability of drawing a card with either a 5 or a 6 of any suit? (Hint: Count the total number of 5s and 6s.)

let $x = 5 \text{ or } 6$

$$P(x) = 8/52 \quad P(x=5 \text{ or } x=6) = P(5) + P(6) = 4/52 + 4/52 = 8/52$$

7. What are the odds of drawing a card from the suit of hearts?

this does not ask for a probability.

hearts : hearts (complement)

ODDS 13 : 39

probability $13/52$

Simple Probability - Examples, cont'd (3)

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Each question is based on rolling a regular 6-sided die.

8. What is the chance of rolling an even number?

$$\text{let } x = 2 \text{ or } 4 \text{ or } 6 \\ \text{then } P(x) = 3/6 = 50\%$$

9. What are the odds of rolling a number divisible by 3?

$$\text{let } x = 3 \text{ or } 6 \\ \text{then } P(x) = 2/6 = 33.\bar{3}\%$$

10. What is the probability of *not* rolling a 1?

Alternative solution:

$$P(\bar{x}) = 1 - P(x) \\ = 1 - 1/6 \\ = 5/6 = 83.\bar{3}\%$$

$$\text{let } x = 1 \\ \text{then } P(\bar{x}) = 5/6 = 83.\bar{3}\%$$

11. Candice and Lior work in the same office. Two people will be randomly selected to attend a conference. If there are 10 people working in the office, what is the probability that both Candice and Lior will be chosen? (**Hint:** Use your work with combinations to find the total ways 2 people can be chosen out of 10. Then count the combinations that include Candice and Lior. Use the information to write the probability ratio.)

Does order matter? No
ways to choose 2 from 10:

$${}_{10}C_2 = \frac{10!}{[8! \cdot 2!]} \\ = \frac{10 \times 9 \times 8!}{2 \times 8!} \\ = 45$$

then let $x = C \& L$

alternative solution:

$$P(L \& C \text{ or } C \& L) \\ = \frac{1}{10} \cdot \frac{1}{9} + \frac{1}{10} \cdot \frac{1}{9} \\ = \frac{1}{90} + \frac{1}{90} \\ = 2/90 \\ = 1/45 \\ = 2.2\%$$

Then let $x = 684$

$$P(x) = 1/45 = 2.29\%$$

$$= 2.29\%$$

12. **Explain** Kayla went to the Pizza Pie Café for lunch and ordered the 2-topping special. She made this comment to a friend: "If you order a 2-topping pizza and the toppings are chosen at random, you will have a 1 out of 28 chance of getting a pepperoni and pineapple pizza."

- a. Assume Kayla is correct. How many different toppings are available at the Pizza Pie Café?

8 Topping

- b. How do you know your answer makes sense? Explain your thinking.

In this problem, we are given the answer.

Toppings n

$$nC_2 = 28$$

$$\frac{n!}{2!(n-2)!} = 28$$

$$\frac{n \times (n-1) \times (n-2)!}{2 \times (n-2)!} = 28$$

$$n(n-1) = 56$$

$$n^2 - n = 56$$

$$n^2 - n + 1/4 = 56 + 1/4$$

$$\sqrt{(n - 1/2)^2} = \sqrt{56.25}$$

$$n = 1/2 \pm 7.5$$

$$n = 8 \text{ or } -7$$

Types of Events

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Dependent Events - Two or more events are dependent if their outcomes influence each other.

For Example:

Suppose you have 6 balls in a hat, 3 are red, 2 yellow, and 1 blue.

Now further suppose you are interested in drawing 2 red balls.

The first time you draw, you have a $3/6$ chance of getting a red ball.

Now, unless you put the ball back in the hat ("replace" it), the second time you draw, you have a $2/5$ chance of getting a red ball.

Therefore, your 1st and 2nd draw are Dependent Events.

Independent Events - Two or more events whose outcomes do not influence each other.

For Example:

Suppose again you have 6 balls in a hat, 3 are red, 2 yellow, and 1 blue.

And further suppose you are interested in drawing 2 red balls.

The first time you draw, you have a $3/6$ chance of getting a red ball.

Now, before your 2nd draw, **replace** the red ball, i.e., put it back in the hat.

Then when you draw the second time, you still have a $3/6$ chance of getting a red ball.

Therefore, your 1st and 2nd draw are Independent Events.

Finally:

The **Dependent Events** were executed *without replacement*.

The **Independent Events** were executed *with replacement*.

Compound Probability

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when we ask **AND**

COMPOUND PROBABILITY

Compound probability is the likelihood of several events happening at once or one after the other. Compound probability is found by multiplying the probabilities of individual events.

Compound events can be **independent or dependent**. Two events are independent when one event doesn't impact the other one. Examples of independent events are flipping a coin more than once or rolling more than one die at a time.

FINDING THE PROBABILITY OF INDEPENDENT EVENTS

Example If you roll two dice, what is the probability of rolling two 6s?

Step 1

Find the probability of the first event.

The probability of rolling a 6 on one of the dice is $\frac{1}{6}$.

The probability of getting two 6s is $\frac{1}{36}$, or 1 out of 36.

Step 2

Find the probability of the second event.

The probability of rolling a 6 on the other die is also $\frac{1}{6}$.

Step 3

Multiply the probabilities.

$$\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

let $x = 2$

$$P(2 \text{ AND } 2)$$

$$= P(2) \cdot P(2)$$

$$= \frac{1}{6} \cdot \frac{1}{6}$$

$$= \frac{1}{36}$$

Two events are dependent when the first event affects the second event.

FINDING THE PROBABILITY OF DEPENDENT EVENTS

Example There are 10 marbles in a bag: 7 are red and 3 are blue. Lisa draws a marble out of the bag. Without replacing the first marble, she draws a second marble from the bag. What is the chance that both marbles are blue?

Step 1

Find the probability of the first event.

Step 2

Assume Lisa has drawn out a blue marble. Now there are 9 marbles left and 2 are blue. Find the probability of the second event.

Step 3

Multiply the probabilities.

$$\frac{3}{10} \times \frac{2}{9} = \frac{6}{90} = \frac{1}{15} \approx 7\%$$

The probability of drawing 2 blue marbles is 1 out of 15, or about 7%.

let $x = \text{Blue}$

$$P(X \text{ AND } X \text{ w/o Replacement})$$

$$= P(X) \cdot P(X|X)$$

$$= \frac{3}{10} \cdot \frac{2}{9}$$

$$= \frac{6}{90}$$

$$= \frac{1}{15}$$

$P(X|X)$ means prob of X given another X already occurred

$$\frac{3 \text{ blue}}{10 \text{ marbles}} = \frac{3}{10}$$

$$\frac{2 \text{ blue}}{9 \text{ marbles}} = \frac{2}{9}$$

If Lisa replaces the marble before drawing out the second marble, the events would be independent. Do you see why? The number of blue and total marbles would not change.

If she replaces the marble, the probability is $\frac{3}{10} \times \frac{3}{10} = \frac{9}{100}$, or 9%.

$$\begin{aligned} P(X \text{ AND } X) &= P(X) \cdot P(X) \\ &= \frac{3}{10} \cdot \frac{3}{10} \\ &= \frac{9}{100} \end{aligned}$$

$$\begin{aligned}
 & \quad \quad \quad 100 \quad \quad 100 \\
 & = \frac{9}{100} \\
 & = 9\%
 \end{aligned}$$

Compound Probability - Examples

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when we ask **OR**

Some problems can be solved by adding probabilities. Mathematicians sometimes call these either/or probabilities.

FINDING AN EITHER/OR PROBABILITY

Example If you roll 2 dice, what is the probability of rolling a 7?

When you roll 1 die, there are 6 outcomes.

When you roll 2 dice, there are 36 outcomes: $6 \times 6 = 36$.

Count the ways to make 7.

There are 2 ways using 1 and 6:

$1 + 6$ and $6 + 1$

The probability is $\frac{2}{36}$.

There are 2 ways using 2 and 5:

$2 + 5$ and $5 + 2$

The probability is $\frac{2}{36}$.

There are 2 ways using 3 and 4:

$3 + 4$ and $4 + 3$

The probability is $\frac{2}{36}$.

+	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12
7	8	9	10	11	12	

Depending on what 2 numbers you get, the probability could be $\frac{2}{36}$ or $\frac{2}{36}$ or $\frac{2}{36}$.

The word **or** tells you to add. $\frac{2}{36} + \frac{2}{36} + \frac{2}{36} = \frac{6}{36} = \frac{1}{6}$

The chart proves the answer. The probability of rolling a 7 is **6 out of 36**, or $\frac{1}{6}$.

ways to get a 7

1, 6 6, 1
2, 5 5, 2
3, 4 4, 3

$$\begin{aligned}
 P(1 \& 6) &= \frac{1}{6} \cdot \frac{1}{6} \\
 \text{or } P(2 \& 5) &+ \frac{1}{6} \cdot \frac{1}{6} \\
 \text{or } P(3 \& 4) &+ \frac{1}{6} \cdot \frac{1}{6} \\
 \text{or } P(4 \& 3) &+ \frac{1}{6} \cdot \frac{1}{6} \\
 \text{or } P(5 \& 2) &+ \frac{1}{6} \cdot \frac{1}{6} \\
 \text{or } P(6 \& 1) &+ \frac{1}{6} \cdot \frac{1}{6} \\
 &= \frac{1}{36} (6) \\
 &= \frac{1}{6}
 \end{aligned}$$

Compound Examples

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Solve.

1. When you flip a coin 3 times, what is the probability of getting 3 heads?

$$\begin{aligned}P(H \& H \& H) &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \\&= \frac{1}{8} \\&= 12.5\%\end{aligned}$$

2. What is the chance of rolling two 6-sided dice and getting a 5 on both dice?

$$\begin{aligned}P(5 \& 5) &= \frac{1}{6} \cdot \frac{1}{6} \\&= \frac{1}{36}\end{aligned}$$

3. A magician has a deck of 20 cards numbered from 1 to 20. He asks a volunteer to draw 2 cards. What is the probability that both cards are greater than 10?

$$\# \text{ cards} > 10 = 10$$

$$\text{Let } x = \text{the card} > 10$$

$$\begin{aligned}P(x \& x) &= P(x) \cdot P(x|x) \\&= \frac{10}{20} \cdot \frac{9}{19} \\&= \frac{90}{380}\end{aligned}$$

$$\begin{aligned}
 &= 90/380 \\
 &= 9/38 \\
 &= 23.7\%
 \end{aligned}$$

4. Suppose you roll one 6-sided die. What is the probability of rolling either a 2 or a 3?

$$\begin{aligned}
 P(2 \text{ or } 3) &= P(2) + P(3) \\
 &= 1/6 + 1/6 \\
 &= 2/6 \\
 &= 1/3 \\
 &= 33.3\%
 \end{aligned}$$

5. Jennifer enrolls in 5 classes: Art, Biology, History, Math 1, and Business Math. If her schedule is assigned randomly, what is the probability that her first 2 classes will be math classes?

$$\begin{aligned}
 &P(M1 \& BM \text{ or } BM \& M1) \\
 &= (1/5)(1/4) + (1/5)(1/4) \\
 &= 1/20 + 1/20 \\
 &= 2/20 = 1/10 = 10\%
 \end{aligned}$$

6. A 3-person committee will be selected from a group of 5 men and 5 women. If names are

6. A 3-person committee will be selected from a group of 6 men and 6 women. If names are drawn at random, what is the probability that all 3 people on the committee will be women?

$$\begin{aligned}
 & P(w \& w \& w) \\
 &= P(w) \cdot P(w|w) \cdot P(w|w \& w) \\
 &= \frac{6}{12} \cdot \frac{5}{11} \cdot \frac{4}{10} \\
 &= \frac{120}{1210} \\
 &= 9.9\%
 \end{aligned}$$

7. Kim buys 10 raffle tickets. Later she finds out that her husband bought 20 tickets. If there are 300 tickets sold in all, what is the chance that either she or her husband will win the prize?

$$\begin{aligned}
 P(\text{win}) &= \frac{30}{300} \\
 &= \frac{1}{10}
 \end{aligned}$$

8. A couple has 3 children. Assume there is a 50% chance of having either a boy or a girl. What is the probability that the couple has 2 boys and 1 girl? (**Hint:** You could make an organized list of the possibilities.)

$$P(\text{BBG or BGB or GBB})$$

1 2 3

$$\begin{aligned}
 & P(BBG \text{ or } BGB \text{ or } GBB) \\
 &= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 \\
 &= \frac{1}{8} (3) \\
 &= 3/8 \\
 &= 37.5\%
 \end{aligned}$$

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Using Probability for Predictions

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BASING PROBABILITY ON DATA

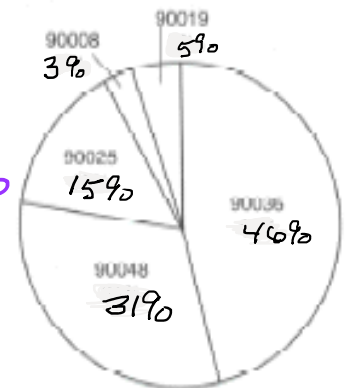
Rob's Roofing Company offers area homeowners a free estimate for a new roof. The company tracks which zip codes yield positive responses to the special offer. The information is shown on the graph to the right.

Example 1 From the names of the homeowners who accepted a free estimate, one name is chosen randomly to win a prize. What is the probability that the name will be from the 90036 zip code area?

The graph represents all positive responses to the promotion. The highest percent on the graph is 46%, so there is a 46% chance of drawing a name from zip code 90036.

The probability is 46%.

Percent of Positive Responses to Free Offer by Zip Code



Example 2 What is the chance that the name will not be from the 90036 zip code?

Subtract: $100\% - 46\% = 54\%$

Since 54% is greater than 46%, it is more likely that the name will be drawn from a zip code other than 90036.

The chance is 54%.

$$1 - P(90036) = 54\%$$

Example 3 If the employees make 220 calls to the 90025 zip code, how many positive responses will they probably get?

$$15\% \text{ of } 220 = 33 \text{ people}$$

You can reasonably conclude that Rob's Roofing Company will continue to have 15% success in the 90025 zip code.

Find 15% of 220.

$$220 \times 0.15 = 33$$

Of the 220 calls, 33 responses will probably be positive.

When given a set of collected data, you can use your knowledge of probability to draw conclusions about what will likely happen.

This concept of predicting is the segue to Statistical Hypothesis Testing in which researchers suggest expected results.

Experimental Probability

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